

# Data Sharing Questionnaire

This questionnaire is about providing respondents recommendations on how to organizationally structure their data ecosystem projects. Additionally, the questionnaire provides a standardized language to help describe/explain data ecosystem projects. No prior knowledge of data ecosystems is required by respondents.

This questionnaire is intended for IT teams responsible for such projects, and takes around 25-30 minutes to complete.

The questionnaire consists of 5 sections:

- **Data:** this is about the characteristic of the data you are planning to work with in this ecosystems.
- **Organization:** this is about organizational part of the data ecosystems, so more about the underlying motivation and etc.
- **Ecosystem complexity:** This part is about exploring the contextual complexity of the project
- **Technical:** this sections dives into understanding the technical dynamics of the data ecosystem.
- **Governance:** this section is about exploring the how the organization wants the data governance to be like in the ecosystem.

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## Section 1. Meta Dimension: Data

*This section is about the data itself. What the data is, what state it is in when it leaves your side, where it came from, how sensitive it is and how often it has to be refreshed all put limits on how it can be shared and on what has to be arranged around it. Answer for the data in this specific project rather than for your organization as a whole.*

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### **Q1. Which of the following categories describe the data the project shares, or will share? (tick all that apply, or write in another)**

- **Process data** --> data from organizational processes, research or development (for example, production line logs, maintenance records, R&D test results)
  - **Product data** --> data from smart products or sensors, sometimes called Big Data (for example, vibration readings from a pump, GPS traces from a truck fleet, machine telemetry)
  - **Environment data** --> data about the surroundings (for example, weather forecasts, air quality readings, soil or terrain maps)
  - **Customer data** --> data specifically about customers (for example, order history, contact details, subscription or usage behaviour)
  - **Metadata** --> data that describes another dataset (for example, a data catalogue entry, column definitions, the owner and timestamp of a file)
  - **I am not sure / not decided yet** --> this still has to be decided, or I do not know it yet
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**Q2. In which state is the data at the moment it is shared? (tick all that apply, or write in another)**

- Raw data --> exactly as received from the source (for example, unfiltered sensor readings, untouched machine log files)
  - Modified data --> data that has been preprocessed (for example, duplicates removed, missing values filled in, units standardized, labels added)
  - Enriched data --> the original data plus extra information from outside the original source (for example, sales figures combined with regional demographics, sensor readings joined with weather data)
  - Processed data --> data ready to be analysed by algorithms or ML models (for example, a cleaned feature table, a pre-aggregated dataset)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q3. From which sources does the shared data originate? (tick all that apply)**

- Generated data --> generated by our own operations, sensors and equipment (for example, our own production line, our own machines and vehicles)
  - Acquired data --> obtained from data generated by other organizations (for example, bought from a data provider, received from a supplier under an agreement)
  - Customer provided data --> collected directly from our own customers (for example, forms customers fill in, survey answers, app usage)
  - Freely available data --> data that is openly published and can be used without an agreement (for example, open government datasets, public weather or map data)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q4. Which level of privacy does the data require? (one answer)**

- Public data --> no restrictions apply, and the data may be shared without restriction (for example, published open datasets, public timetables, published environmental readings)
  - Internal data --> the data can be shared within the domain, but requires restrictions when shared outside the domain (for example, internal performance reports, team dashboards)
  - Confidential data --> this level of privacy is intended for organizational secrets, where if leaked, will possibly hurt the organization. (e.g. company records, trade secrets, source code)
  - Sensitive data --> the data contains personally identifiable information, information related to private information of individuals. (for example, employee records, patient data, customer personal data under GDPR)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q5. How frequently does the data have to be updated once it is shared? (one answer)**

- Static --> no updates are required after sharing (for example, a one-off historical export, a fixed reference table)
  - Up to date --> repeated updates on a schedule (for example, a daily or weekly batch upload)
  - (near) Real time --> a live stream or continuous feed (for example, sensor telemetry pushed the moment it is measured)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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## **Section 2. Meta Dimension: Organization**

*This section is about why the data is shared and with whom: what the organization seeks from the exchange, which parties take part in it, and what is given or received in return. Answer for this specific project rather than for your organization as a whole.*

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**Q6. What motivates the sharing of this data? (tick all that apply, or write in another)**

- Economic --> money or efficiency (for example, new revenue from selling data, cost savings, better forecasting)
  - Social and environmental --> to contribute to social or environmental goals (for example, sustainability reporting, emissions tracking, corporate social responsibility programmes)
  - Legal --> legislation or a regulator requires the data to be shared (for example, GDPR obligations, EU Data Act, regulator or safety reporting)
  - Cultural --> to improve the culture around data sharing within the organization (for example, breaking down silos between departments, building trust in shared data)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q7. With whom is the data shared, or to be shared? (one answer, or write in another)**

- Intra-organizational --> the data stays inside our own organization (for example, between our production, sales and logistics departments)
  - Same industry --> only with organizations in the same industry (for example, a supplier sharing operational data with a manufacturer)
  - Cross-industry --> across sector boundaries (for example, sharing meteorology data with agriculture or aerospace companies)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q8. What does your organization expect in return for sharing the data? (tick all that apply, or write in another)**

- Financial reward --> a monetary return (for example, a licence fee, a price per dataset)
  - Virtual assets --> digital assets that cannot be directly converted into cash (for example, platform credits, tokens, points)
  - Data in return --> data is received in exchange for the data shared (for example, we share production data, the partner shares market data)
  - Services --> additional or new services from the receiving party (for example, a free analytics dashboard, a maintenance or benchmarking service)
  - (Increased) Reputation --> greater brand awareness, trust or credibility (for example, recognized as a trusted partner, publicity as a frontrunner in the sector)
  - Nothing in return --> no return is expected (for example, the data is shared because a regulator requires it, or it is published openly)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q9. Which form of compensation is expected for the data? (one answer)**

- Free --> the data is shared free of charge (for example, internal sharing between departments, open data, legally mandated reporting)
  - Fixed --> a single, fixed price, paid at once or in instalments (for example, EUR 5,000 for a dataset)
  - Subscription based --> a recurring price every set period (for example, a monthly or yearly access fee)
  - Usage based --> the compensation varies with how much the data is used (for example, a price per API call, per record downloaded)
  - Revenue based --> compensation based on the revenue generated from the shared data (for example, 8% of the revenue generated from the data, or royalties)
  - Hybrid --> a combination of the models mentioned above (for example, a base subscription plus a charge per API call)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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## **Section 3. The contextual complexity of the Project**

*This section is about the complexity and the maturity of the project: how settled its workflows are, how the actors relate to one another, how much is still uncertain about the data and the systems to be connected, and how far the project reaches into the organization. Answer for the situation as it stands now, rather than for the situation you expect to reach.*

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**Q10. Which description fits the operations or workflows of the project? (one answer)**

- Well defined --> the workflows were already in operation before this project, with little room for change (for example, an established production process, a reporting routine that already runs)
  - Still in development --> the workflows are being created for this project and are open to change (for example, a new service being designed, a pilot with no fixed process yet)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q11. Which actors take part in the project? (one answer)**

- Many actors, both inside and outside our organization (for example, our own product development, sales and operations departments together with customers, suppliers, maintenance providers and logistics partners)
  - Mainly actors inside our own organization, with coordination kept internal (for example, production, engineering and IT teams across our own sites)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q12. What does the involvement of the stakeholders require? (one answer)**

- Incentives have to be aligned, decisions agreed, and ownership and responsibilities defined across the stakeholders (for example, negotiating who pays, who is liable, and who decides, between parties that can walk away)
  - The stakeholders are already part of an internal or existing ecosystem, so governance is more direct (for example, decisions can be escalated to a common manager or an existing steering group)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q13. How certain is it which data will be exchanged, and how it is measured? (one answer)**

- The data to be exchanged, and how it is measured, are still uncertain (for example, the fields, units and quality are still to be discovered)
  - Data structures and exchanges exist from before this project, so there is little uncertainty (for example, the schemas and formats are already documented and in use)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q14. What does the integration across the participating systems require? (one answer)**

- Different systems and data formats may have to be accommodated (for example, partners running different ERP systems, data arriving in incompatible formats)

- The integration plan and the data formats are already known and standardized (for example, one agreed schema, systems that already talk to each other)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q15. How many systems, data formats or organizations does the project have to connect? (one answer)**

- A few, all inside our own organization, and already known (for example, two internal systems exchanging one agreed format)
  - Several, inside our own organization but across different domains or departments (for example, the production system, the ERP and the sales system, each with its own format and owner)
  - Many, including systems owned by other organizations outside ours (for example, several partners, each bringing their own systems, formats and standards)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q16. How will the project affect the organization? (one answer)**

- New mechanisms and organizational structures will be put in place (for example, new roles such as data owners or stewards, new decision-making bodies, changed responsibilities)
  - The project is mainly one of system integration and should not affect workflows or the organization (for example, two systems get connected, everyone keeps working the way they did)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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## **Section 4. Meta Dimension: Technical**

*This section is about how the exchange works in practice: the route the data takes between the parties, where the data sits once it is shared, what has to line up on both sides for it to be understood, how access is regulated, what sets an exchange off, and how many parties are involved. Answer for how you want the sharing to work, not only for what is already built.*

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**Q17. By which route is the data exchanged between the participants? (one answer, or write in another)**

- Peer to peer --> the data passes directly from one party to the other, with no intermediary (for example, our system calls theirs directly, or they call an API we run ourselves)
  - Platform based --> the exchange happens through the intermediation of a platform (for example, a data marketplace, a shared portal, a broker that sits between provider and consumer)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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### **Q18. Where does the data reside when it is shared? (one answer)**

- Centralized --> copied into one shared location, from which consumers retrieve it (for example, a central data warehouse, data lake, or company-wide platform)
  - Distributed --> the data stays where it is produced and is accessed at the source through an agreed interface (for example, each domain or organization exposes its own data through a connector or API; nothing is duplicated centrally)
  - Hybrid --> some data is copied into a central store, other data stays at the source and is accessed there
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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### **Q19. Which levels of interoperability does the exchange require between the parties? (tick all that apply)**

- Legal interoperability --> compatible legal frameworks, standards and regulations (for example, matching contract clauses, GDPR-compliant terms, agreed usage policies)
  - Organisational interoperability --> a shared understanding of processes, objectives and resources (for example, agreed roles, responsibilities and escalation paths between the parties)
  - Semantic interoperability --> consistency and interpretability across diverse sources and systems (for example, a shared vocabulary or ontology, so that "customer" or "downtime" means the same thing on both sides)
  - Technical interoperability --> the same technical data exchange across systems (for example, matching formats and protocols, JSON over REST, OPC UA, agreed schemas)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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### **Q20. On what basis is access to the data granted? (one answer)**

- Agreement based --> access terms are defined between you and the other party (for example, a bilateral data sharing contract, an NDA with data clauses)
  - Application based --> access is regulated through those who use the specific application (for example, only users logged into the shared dashboard can see the data)
  - Trust based --> access is granted on the basis of trust (for example, known partners are granted access without a formal contract or technical gate)
  - Open --> no access restrictions apply, and anyone may access the data (for example, published as open data)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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### **Q21. What triggers an exchange of data? (one answer, or write in another)**

- On demand --> sharing is triggered when the consumer asks for the data (for example, the consumer queries the API when they need it)

- Event based --> triggered by an event specified by the provider (for example, end of day, the dataset reaching a certain size, an alarm or threshold being crossed)
  - Continuous --> sharing data in real time or at regular intervals (for example, a live stream, or an automatic push every 15 minutes)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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## **Q22. Which pattern do the data sharing relationships follow? (one answer)**

- One-to-one --> one party shares data with one other party (for example, one supplier sends data to one manufacturer)
  - One-to-many --> one party shares data with many other parties (for example, one provider publishes a dataset that many consumers use)
  - Many-to-one --> many parties share data with one other party (for example, many suppliers report into one OEM)
  - Many-to-many --> many parties share data with many other parties (for example, an ecosystem where every participant both provides and consumes)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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## **Q23. How dependent are the parties on one another's data? (one answer)**

- Loosely coupled --> each party can continue to operate if the data stops arriving (for example, the shared data improves a forecast, but the process still runs without it)
  - Tightly coupled --> the parties depend on each other's data to operate (for example, a production schedule that cannot be made without the partner's live data)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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# **Section 5. Meta Dimension: Governance**

*This section is about control over the data once it has been shared: who decides what may be done with it afterwards, and who holds the authority to set the rules for it and see that they are followed. These questions are often still open early in a project; please use the "I am not sure / not decided yet" option rather than guessing.*

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## **Q24. Once the data has been shared, who retains control over what may be done with it? (one answer)**

- Sovereign control --> the provider keeps control over how the data may be used after sharing (for example, usage policies travel with the data, the receiver may only use it for an agreed purpose)
- Transferred control --> the rights of use pass to the party that receives the data (for example, we hand the dataset over and they may use it as they see fit)

- Shared control --> no restrictions are placed on ownership or use, and the parties hold it in common (for example, a dataset produced jointly by a consortium that every member may use freely)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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**Q25. Who holds the authority to set the rules for the shared data and to ensure that they are followed? (one answer)**

- Decentral units --> each party decides for its own data: whoever provides it sets the rules for it and decides who may access it (for example, each domain or organization sets and enforces the rules for the data it publishes)
  - A central unit --> one unit decides for all the parties, including for data it does not own (for example, a central data office, a data intermediary, a trustee or broker acting on behalf of the parties)
  - Both --> a central body and the parties share the authority (for example, a consortium body or joint steering group sets the common rules, and each party decides within them)
  - I am not sure / not decided yet --> this still has to be decided, or I do not know it yet
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